

Viability of fracture characterization using seismic attributes through seismic physical modeling

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Abstract

Seismic attributes have been widely utilized for fault detection; however, accurately characterizing mesoscale fractures (1/10 to 1/4 wavelength) using these attributes remains challenging, particularly in complex geological setting. To explicitly reveal the effectiveness of seismic attributes in fracture characterization, we constructed a realistic physical model based on the stratigraphic structure of a real shale-gas play. This model incorporates several faults and six fracture zones containing vertically aligned mesoscale fractures of varying fracture densities, embedded in a target layer of 40 m equivalent thickness. Through a comprehensive evaluation of various seismic attributes, we demonstrate that, while these attributes effectively map faults or macroscale fractures, they are less reliable for characterizing mesoscale fractures. Although mesoscale fractures induce anomalies in seismic amplitude and attributes, the magnitude of these anomalies does not correlate with the fracture density. To address this limitation, we introduce the amplitude at the dominant frequency (ADF) attribute, designed to characterize fracture density in thin beds with mesoscale fractures. A strong relationship is observed between fracture density and the ADF attribute, with a correlation coefficient of 0.8067. Thus, the ADF attribute serves as a valuable tool for interpreting seismic data in the context of mesoscale fracture density.

Keywords: fractures characterization; seismic attribute; spectral decomposition; physical model

1. Introduction

Fractures play a critical role as migration pathways and provide additional storage space for oil and gas, making them essential for exploration and development (Curtis 2002). Fractures vary in scale and are commonly categorized as macroscale, mesoscale, and microscale (MacBeth 1995). Macroscale fractures typically span tens of meters to kilometers (exceeding 1/4 wavelength), mesoscale fractures range from a few centimeters to several meters (1/10 to 1/4 wavelength), while microscale fractures occur at the grain scale of

rock, with dimensions significantly smaller than a 1/10 wavelength.

Seismic attributes, such as coherence, variance, curvature, and ant-tracking, have been widely employed to facilitate the detection of faults and macroscale fractures (Roberts 2001; Chopra 2002; Pedersen *et al.* 2002; Al-Dossary & Marfurt 2006; Barnes 2007; Mendrii & Dovbnich 2014; Ali *et al.* 2021; Chen *et al.* 2023; Jing *et al.* 2023; Dou *et al.* 2024). Several studies have demonstrated the potential of these attributes to identify subtle faults and closely spaced fracture

clusters (Neves *et al.* 2004; Khromova *et al.* 2011). For instance, Hunt *et al.* (2010) found that the curvature attribute derived from 3D seismic volumes correlates strongly with fracture density, enabling quantitative prediction of fracture density in the Nordeg area. To improve the accuracy and efficiency of fault detection, machine learning-based methods have also been explored for extracting fault information from seismic data (Xiong *et al.* 2018; Wu *et al.* 2019; Bi & Wu 2021; Wang & Wang 2022; Jing *et al.* 2023). Despite these advancements, seismic attributes are generally considered to be most effective in predicting faults and macroscale fractures (Sliz & Al-Dossary 2014; Ding *et al.* 2023).

Velocity and amplitude variations with azimuth (VVAZ or AVAZ) from wide-azimuth P-wave seismic data have proven useful for characterizing aligned microscale fractures (Williams-Stroud *et al.* 2013; Bouchaala *et al.* 2023). Attributes related to azimuthal anisotropy, such as amplitude and velocity, are extracted from pre-stack azimuth seismic gathers and used to infer mesoscale fracture density and orientation (Rüger & Tsvankin 1997; Rüger 1998; Li 1999; MacBeth & Li 1999; Bakulin *et al.* 2000; Cho *et al.* 2013; Takougang *et al.* 2019; Zhang *et al.* 2020; Li *et al.* 2022; Chen & Innanen 2023). However, pre-stack seismic gathers with azimuthal information are not always available in some oil fields, limiting the technique's potential for fracture characterization (Hart *et al.* 2002; Shen *et al.* 2002; Cho *et al.* 2013).

Several studies have demonstrated that spectral decomposition enables geophysicists to extract geological insights from seismic data by uncovering hidden frequency information typically indiscernible in conventional seismic datasets (Partyka *et al.* 1999; Nikoo *et al.* 2016). Over the past few decades, spectral decomposition has emerged as a valuable tool for seismic interpretation, proving effective in various applications such as estimating thickness in thin beds (Partyka *et al.* 1999; Sun *et al.* 2009; Saadatinejad *et al.* 2011; Liu *et al.* 2022a), stratigraphic delineation (Marfurt & Kirilin 2001; Liu *et al.* 2011; Bonar & Sacchi 2013; Li *et al.* 2016; Ma *et al.* 2019; Liu *et al.* 2022b), reservoir characterization (Li & Zheng 2008; Sun *et al.* 2010; Li *et al.* 2011; Torrado *et al.* 2014; Ghosal & Juhlin 2018), and fluid identification (Castagna *et al.* 2003; Sinha *et al.* 2005; Lu & Li 2013; Krylova & Goloshubin 2017; Ahmad *et al.* 2019; Xue *et al.* 2020; Wang 2021; Miao *et al.* 2024). Lyu *et al.* (2020) developed an effective tool, called multispectral coherence, by employing various spectral decomposition methods to highlight geological features. Xie *et al.* (2022) proposed an improved ant-tracking workflow, where ant-tracking is performed on low-, medium-, and high-frequency seismic data cubes to enhance the accuracy of fracture detection. Nevertheless, there has been limited research on the detection of mesoscale fractures in thin beds using spectral decomposition techniques.

In this investigation, we first evaluate the viability of several commonly used seismic attributes for characterizing

faults and mesoscale fractures. A physical model is designed based on the stratigraphic features of a real shale-gas play in the Southern Sichuan Basin. 3D seismic data is acquired from the physical model, which is submerged in a water tank. Attributes such as coherence, variance, curvature, and ant-tracking are calculated from the seismic dataset of the physical model, and the results are compared with the known faults and fracture zones. Additionally, we introduce the amplitude at the dominant frequency (ADF) attribute within the first tuning period to predict the density of mesoscale fractures in thin beds. This study provides valuable insights and a practical tool for fracture characterization.

2. Construction of physical model, seismic data acquisition, and processing

2.1. Physical model construction of a shale-gas play

The Longmaxi formation in the Sichuan Basin is a stratum from the early Silurian period that contains graptolites and ranges in depth from 3800 to 3200 m. The lower Longmaxi formation is a high-quality shale-gas reservoir with high organic matter content. In the study area, the thickness of the lower Longmaxi formation varies from 5 to 18 meters (Yang *et al.* 2022). Drilled wells reveal that 80% of gas production is associated with the development of natural fractures (Yang *et al.* 2019).

To assess the effectiveness of seismic attributes in characterizing faults and fractures, we construct a realistic physical model of shale that incorporates both macroscale and mesoscale fractures. The design of this physical model was informed by interpreted stratigraphic data from a field 3D seismic survey. The physical model was scaled at 1:10 000, with dimensions of 1300 mm in length and 1000 mm in width, representing a study area equivalent to 13 000 by 10 000 m. Figure 1a illustrates the seven-layer physical model, which includes faults and fracture zones. From top to bottom, the seven layers in the model are Lower Permian (L1), Longmaxi 2–2 (L2), Longmaxi 2–1 (L3), Longmaxi 1–2 (L4), Longmaxi 1–1 (L5), Ordovician (L6), and Cambrian (L7). The target layer, Longmaxi 1–1 (L5), a shale formation depicted in Fig. 1b, has a total thickness of 4 mm, corresponding to 40 m at field scale, and contains six distinct fractured zones.

Six fractured blocks are embedded within the layer L5, with vertically aligned fractures designed to simulate varying fracture densities. Each fractured block has a thickness of 2 mm, with a width and length of 40 mm, which correspond to field-scale dimensions of 20 × 400 × 400 m. Zone 3–2 has the lowest fracture density, at 28 km^{−1}, while Zone 3–1 has the highest, at 99 km^{−1}. Fracture densities and elastic parameters of the fractured blocks were measured in the laboratory, as shown in Table 1. All model parameters were set to match those of the real target formations.

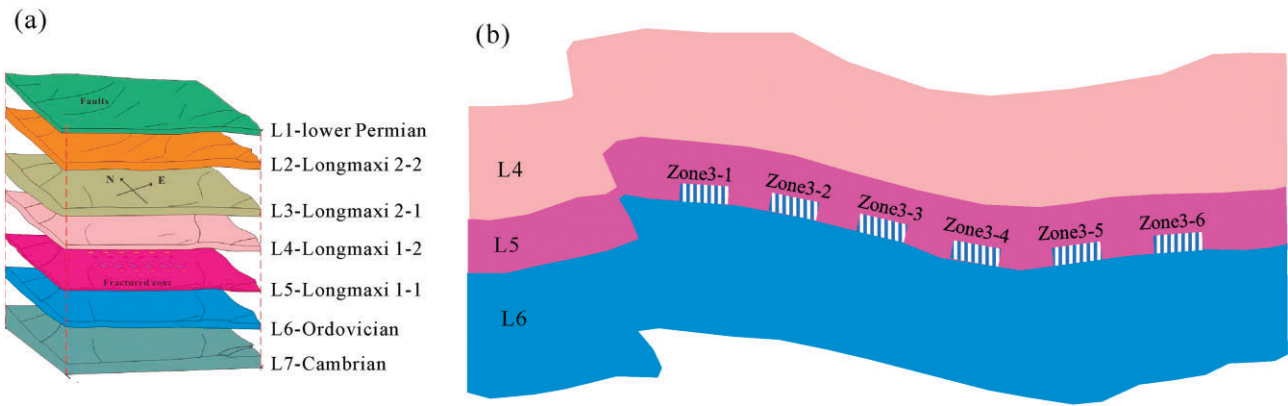


Figure 1. (a) The seven-layer 3D physical model containing faults and fracture zones. (b) The W-E profile of layer L4-L6 with six fractured blocks embedded in the layer L5. The total thickness of the target layer L5 is 4 mm, equivalent to 40 m at field scale.

Table 1. Fracture density and elastic parameters measured in the laboratory for the six fracture blocks. Fracture density represents the number of embedded fracture planes per kilometer.

Fracture zones	True fracture density (km^{-1})	V_p (km/s)	V_s (km/s)	Density (g/cm^{-3})
Zone 3-1	99	3.696	1.762	2.224
Zone 3-2	28	4.024	1.908	2.252
Zone 3-3	33	4.002	1.884	2.249
Zone 3-4	38	3.988	1.888	2.246
Zone 3-5	41	3.944	1.856	2.242
Zone 3-6	47	3.858	1.814	2.238

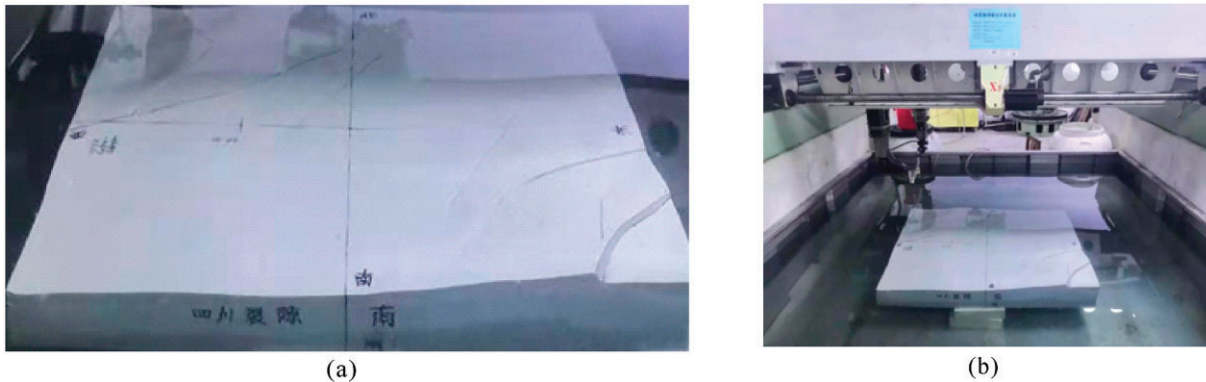


Figure 2. (a) The physical model and (b) the laboratory seismic data acquisition system.

In summary, the target layer L5 of the physical model contains three faults (F1–F3) and six fracture zones (Zones 3–1 to Zones 3–6), each with vertically aligned fractures of varying densities. The faults are classified as macroscale fractures, while the fracture zones are categorized as mesoscale fractures.

2.2. Seismic data acquisition and processing

The 3D seismic dataset was acquired in a laboratory water tank, with the physical model fully submerged. Ultrasonic wave transducers, acting as both source and receiver, were po-

sitioned 0.5 mm below the water surface. Figure 2 presents the constructed physical model along with the laboratory seismic data acquisition system. The laboratory acquisition parameters, outlined in Table 2, were designed based on the 3D seismic acquisition geometries used in the field study area in the Southern Sichuan Basin. The seismic acquisition geometry is depicted in Fig. 3, consisting of 36 receiver lines spaced 280 m apart and 41 shot lines spaced 320 m apart. A total of 17 640 shots and 17 604 geophone stations were distributed across the survey area, yielding a fold coverage of 210.

Table 2. The 3D seismic data acquisition parameters.

Acquisition parameters	Physical model scale	Equivalent field scale
Shot line interval	32 (mm)	320 (m)
Receiver line interval	28 (mm)	280 (m)
Trace interval	4 (mm)	40 (m)
Shot interval	4 (mm)	40 (m)
Number of traces	120	120
Time sampling points	4096	4096
Time sampling interval	1 (ms)	1 (ms)
Minimum offset	20 (mm)	200 (m)
Maximum offset	496 (mm)	4960 (m)
Fold number	210	210
Bin size	2 (mm) × 2 (mm)	20 (m) × 20 (m)
Azimuth ratio	0.8477	0.8477
Dominant frequency	150 kHz	30 Hz

Data processing is crucial for enhancing the signal-to-noise ratio in seismic data and enabling more accurate seismic interpretation. A comprehensive seismic data processing workflow was applied to the acquired 3D seismic dataset. The raw data are significantly contaminated by linear noise at certain frequencies, as well as by multiples. To mitigate these issues, low-frequency linear noise in the shot gathers was suppressed in the frequency-space domain by removing signal components below 20 Hz and 2000 m/s (Fig. 4). Similarly, high-frequency linear noise in the receiver gathers was attenuated by filtering out signal components in the 20–85 Hz range and below 2900 m/s in the frequency-space domain (Fig. 5).

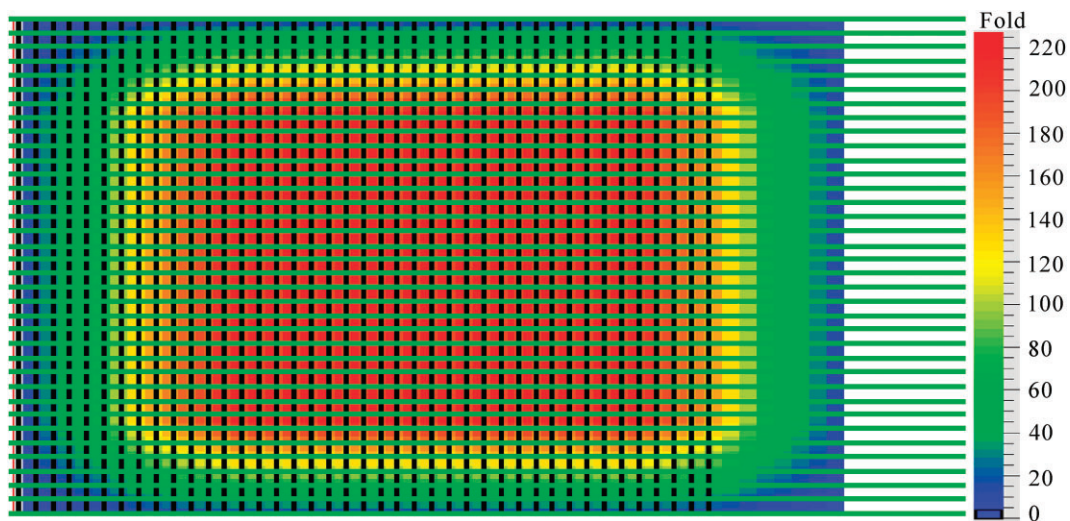
Multiples are attenuated using a combination of Radon transform and deconvolution. The Radon transform is applied to the NMO-corrected CMP gather, where reflections within the range of intercept time of 1900 ms and slowness of 120–1400 ms/km are filtered out. Deconvolution is subse-

quently applied in the shot domain to address internal multiples, with a white noise percentage of 0.5%. Figure 6 compares the CMP gather before and after multiple suppression, showing a significant reduction in multiples and an improvement in velocity analysis.

Data processing can influence seismic amplitude to some extent. Throughout the processing workflow, particular attention is given to preserving true amplitude by carefully selecting parameters, such as those for the Radon transform and deconvolution. As shown in Fig. 6, the AVO variation is well preserved after processing, and the reflection events are not overly flattened. Figure 7 presents a CRP (common reflection point) gather and a stacked seismic profile after processing, both of which display a high signal-to-noise ratio. The reflection events in the CRP gather are well flattened, and the stacked section shows good lateral continuity.

3. Comparison of seismic attributes in multiscale fracture characterization

Seismic attributes such as coherence, variance, and curvature have been widely employed for detecting faults, subtle faults, and fracture swarms (Hunt *et al.* 2009; Hunt *et al.* 2010; Mendrii & Dovbnich 2014). In the physical model used for this study, three faults (F1–F3) and six fracture zones (Zones 3–1 to Zones 3–6) are present. In this section, we assess the effectiveness of several commonly used seismic attributes (coherence, variance, curvature, and ant-tracking) in characterizing these faults and fracture zones using the seismic dataset derived from the physical model. Based on the work of Dorn & Dominguez (2017), structure-oriented filtering, as proposed by Höcker & Fehmers (2002), was first applied to the seismic data to suppress random noise and enhance fault imaging (Dorn *et al.* 2012).

**Figure 3.** Shot locations (black), receiver positions (green), and fold coverage (color bar) for the 3D seismic data acquisition of the physical model.

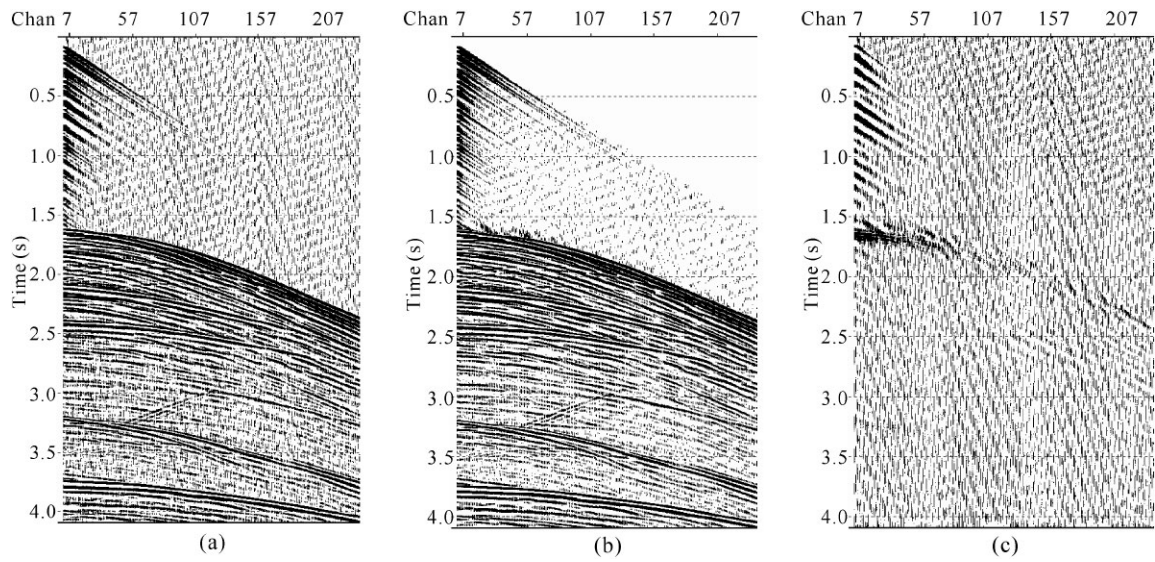


Figure 4. A shot gather (a) before and (b) after the suppression of low-frequency linear noise. (c) The suppressed low-frequency linear noise.

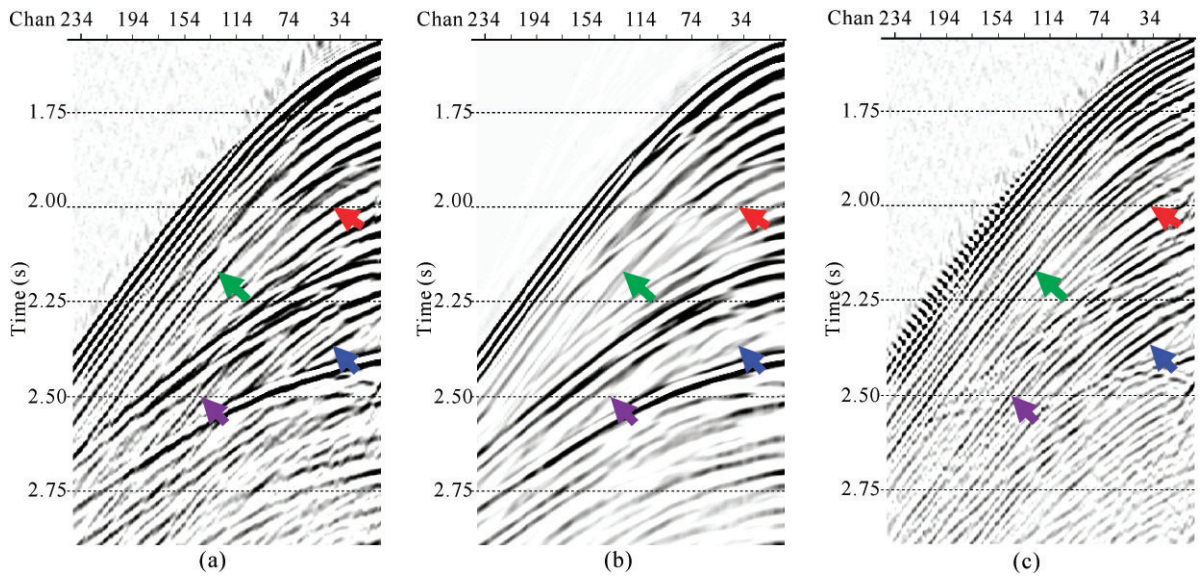


Figure 5. A receiver gather (a) before and (b) after the attenuation of high-frequency linear noise. (c) The attenuated linear noise.

Seismic sections are initially presented to illustrate the amplitude response within the fault and fracture zones. In the seismic section of Inline 275 (Fig. 8a), six fracture zones with varying fracture densities are delineated. While amplitude anomalies are visible within these fracture zones, the correlation between amplitude and fracture density is weak. This is further evident in the horizon slice intersecting the fracture zones, where the zones are highlighted with black squares (Fig. 8b). Although the faults (F1, F2, and F3) are identifiable in the horizon slice, the resolution is insufficient for detailed analysis. Consequently, seismic amplitude alone does not provide a reliable quantitative interpretation of fracture density.

3.1. Coherence attribute

Coherence attributes, which measure the similarity between waveforms, serve as an effective tool for discontinuity detection (Bahorich et al. 1995). The application of coherence attributes has been demonstrated to effectively visualize geological discontinuities such as faults and fractures (Gersztenkorn & Marfurt 1999; Cui et al. 2016; Chopra & Marfurt 2019). Gersztenkorn & Marfurt (1999) presented an eigenstructure-based coherence algorithm. The coherence estimate based on eigenstructure (E_c) can be expressed as follows:

$$E_c = \frac{\lambda_1}{\text{Tr}(\mathbf{C})} = \frac{\lambda_1}{\sum_{j=1}^J \lambda_j}, \quad (1)$$

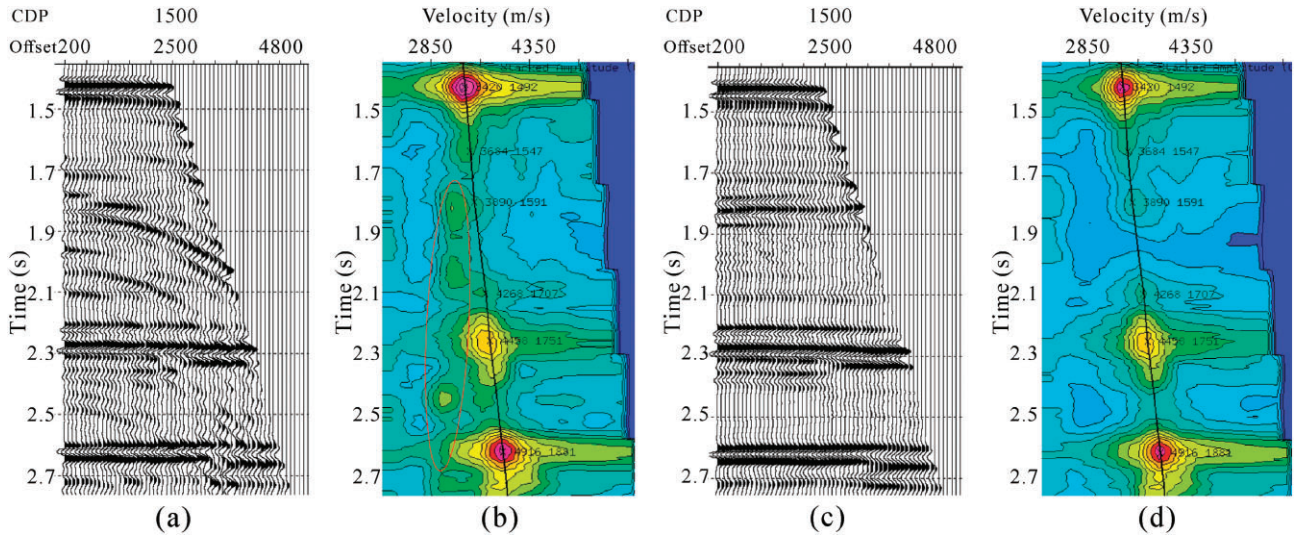


Figure 6. A CMP gather (a) before multiple attenuation and (b) its corresponding velocity spectrum. The CMP gather (c) after multiples attenuation and (d) its corresponding velocity spectrum.

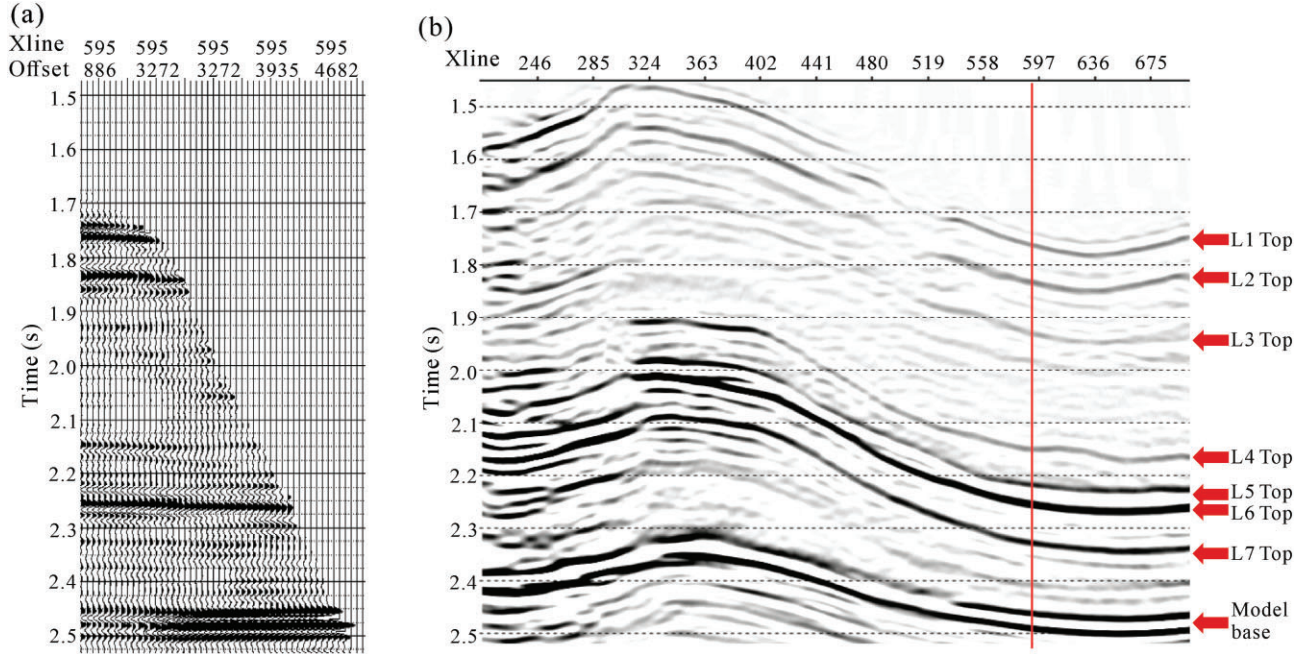


Figure 7. (a) A CRP gather after processing. (b) A seismic profile after pre-stack time migration, with the red line indicating the location of the CRP gather.

$$Tr(C) = \sum_{j=1}^J \sum_{n=1}^N d_{nj}^2 = \sum_{j=1}^J \lambda_j \quad (2)$$

$$D = \begin{bmatrix} d_{11} & d_{12} & \cdots & d_{1J} \\ d_{21} & d_{22} & \cdots & d_{2J} \\ \vdots & \vdots & \ddots & \vdots \\ d_{N1} & d_{N2} & \cdots & d_{NJ} \end{bmatrix}, \quad (3)$$

where d_{nj} is the amplitude of the n th sample of the j th trace, $C = D^T D$ is the covariance matrix of the matrix D , $Tr(C)$ is the numerical trace of the covariance matrix C , λ_j is the j th eigenvalue of C , and λ_1 is the largest eigenvalue of λ_j .

Using the eigenstructure-based coherence algorithm, we extract the coherence attribute from the physical modeling seismic data. The coherence attribute is merged with the seismic section, as shown in Fig. 9a, and the corresponding horizon slice is shown in Fig. 9b. Low coherence anomalies are observed for faults F1–F3, which are better characterized

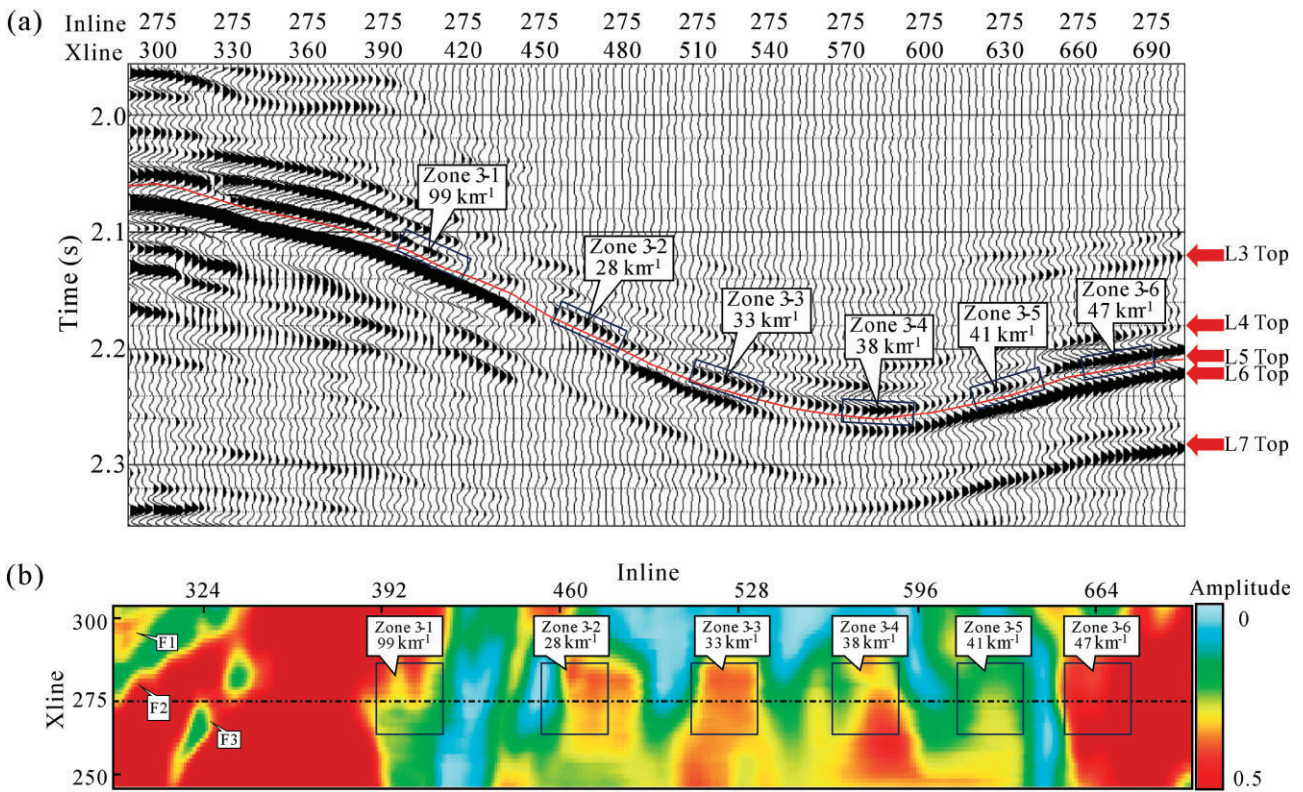


Figure 8. (a) Seismic profile of Inline 275. (b) Horizon seismic slice of along the top of layer L5. The black squares indicate fractured zones.

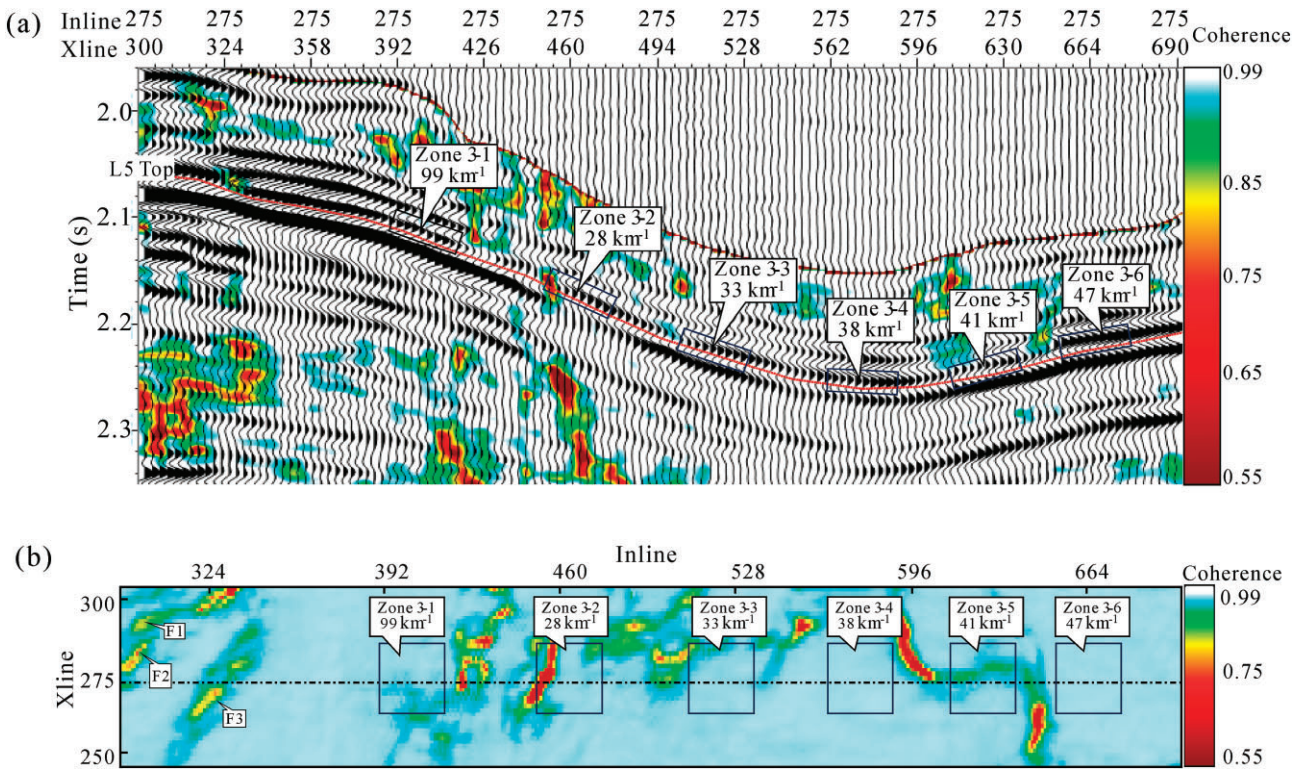


Figure 9. (a) Merged coherence attribute and seismic profiles of Inline 275. (b) The coherence slice along the top of layer L5. The black squares indicate fractured zones.

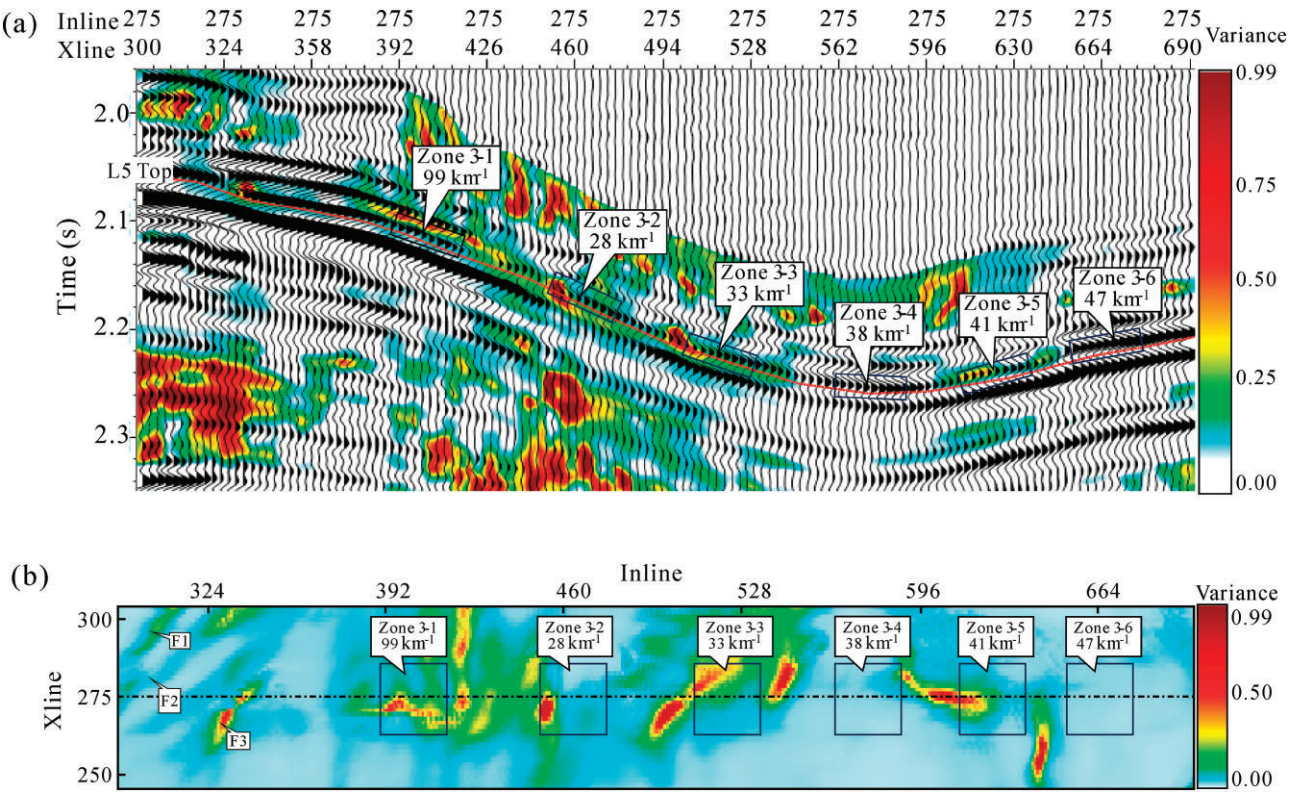


Figure 10. (a) Merged variance attribute and seismic profiles of Inline 275. (b) The variance slice along the top of layer L5. The black squares indicate fractured zones.

than using seismic amplitude (Fig. 9b). Several coherence anomalies are found near the edges of the fracture zones. However, no coherence anomalies are observed within the fracture zones, which may be due to the relatively small scale of the fractures and the effects of thin-bed tuning.

3.2. Variance attribute

The variance attribute, which measures the degree of dissimilarity, is frequently used to detect changes in seismic data resulting from variations in subsurface geological conditions (Sukmono et al. 2006). The variance σ_i^2 of the i th seismic trace d_i can be expressed as follows (Barnes 2016):

$$\sigma_i^2 = \frac{1}{N} \sum_{k=1}^N (d_{ki} - \mu_i)^2, \quad (4)$$

where k is the sample number, N is the total number of sample points, and μ_i and σ_i represent the mean value and the standard deviation of the i th seismic trace d_i , respectively.

Using the variance algorithm, we extract the variance attribute (Daber et al. 2011) from the physical modeling seismic data. The variance attribute is merged with the seismic section, as shown in Fig. 10a, and the corresponding horizon slice is shown in Fig. 10b. Variance traces away from faults or fracture zones exhibit low variance coefficients, whereas

discontinuities such as faults and fracture zones result in high variance coefficients (Liao et al. 2020). Faults (F1, F2, and F3) are visible in the variance attribute. High variance anomalies are observed in fracture Zones 3-1, 3-2, 3-3, and 3-5 (Fig. 10), indicating that the variance attribute can characterize the fracture characteristics of these zones.

3.3. Curvature attribute

The curvature attribute can be employed to delineate subsurface geological structures such as folds, faults, and fracture swarms (Roberts 2001; Mendrii & Dovbnich 2014; Somasundaram et al. 2017). Existing investigations have demonstrated that the most positive curvature and most negative curvature attributes provide superior interpretation of subtle fault details (Roberts 2001; Chopra & Marfurt 2007; Hunt et al. 2010).

Roberts (2001) defines a surface $z(x, y)$ as follows:

$$z(x, y) = ax^2 + by^2 + cxy + dx + ey + f, \quad (5)$$

where x and y represent planar coordinates, while z denotes depth or time. The coefficient f represents the depth or time of the horizon at $x = 0, y = 0$. The remaining coefficients (a, b, c, d and e) are derived from the first- and second-order partial derivatives with respect to x and y .

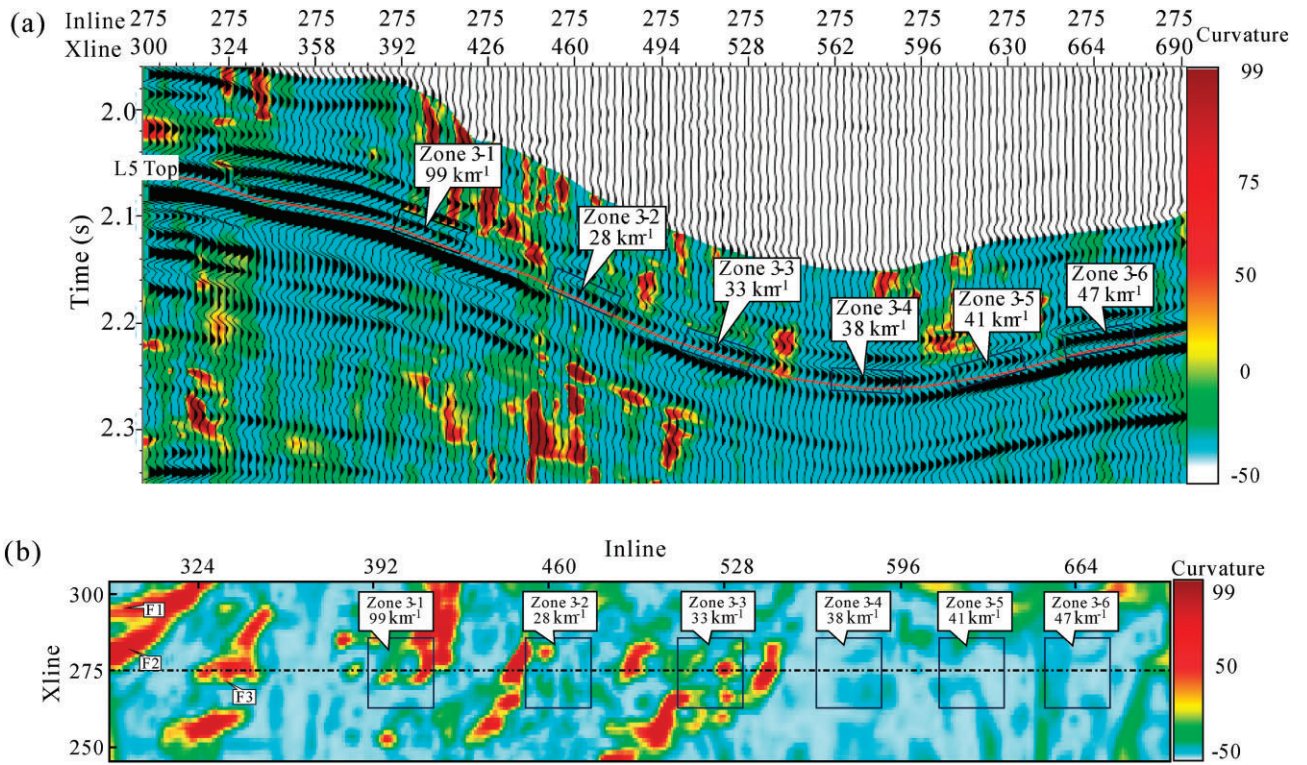


Figure 11. (a) Merged most positive curvature attribute and seismic profiles of Inline 275. (b) The most positive curvature slice along the top of layer L5. The black squares indicate fractured zones.

Then the most positive curvature k_{pos} is calculated as

$$k_{pos} = (a + b) + [(a - b)^2 + c^2]^{1/2}, \quad (6)$$

and the most negative curvature k_{neg} is calculated as

$$k_{neg} = (a + b) - [(a - b)^2 + c^2]^{1/2}. \quad (7)$$

Using the curvature algorithm, we extract the most positive curvature and the most negative curvature attributes from the physical modeling seismic dataset. The curvature attributes are merged with seismic section, as shown in Figs 11a and 12a, and the corresponding horizon slices are shown in Figs 11b and 12b, respectively. High curvature, whether positive or negative, facilitates the delineation of geological structures, including folds, faults, and fractures (Somasundaram *et al.* 2017). Faults (F1, F2, and F3) can be clearly characterized using the most positive curvature. Anomalies in the most positive curvature are observed in Zone 3-1 and Zone 3-3, and those of the most negative curvature are observed in Zone 3-2. Curvature anomalies are not prominently visible in other fractured zones. The most negative curvature attribute performs poorly in characterizing fracture zones.

3.4. Ant-tracking attribute

The ant-tracking attribute is a highly effective edge enhancement technique for identifying faults and swarms of fractures

(Li *et al.* 2013). Pedersen *et al.* (2002) employed the ant colony optimization technique, a form of swarm intelligence algorithm, to link discontinuous similarity events and reduce certain anomalous noise, thereby enhancing the desired fault or fracture characteristics. The general ant-tracking workflow consists of three phases (Daber *et al.* 2011): (i) seismic data conditioning (e.g. structural smoothing filtering) to reduce noise and enhance fault images, (ii) extracting seismic discontinuity volumes, such as the variance attribute volume, to improve the spatial representation of discontinuities in the seismic data, and (iii) extracting the ant-tracking volume from the seismic discontinuity cubes to enhance the fault or fracture. This enhancement is achieved by placing artificial ants at starting points on a seismic discontinuity volume to identify fault or fracture regions.

Following the ant-tracking workflow, we extract the ant-tracking attribute from the physical modeling seismic data. The ant-tracking attribute is merged with the seismic section, as shown in Fig. 13a, and the corresponding horizon slice is shown in Fig. 13b. A higher value in the ant-tracking attribute indicates a greater likelihood of identifying the presence of faults or fractures. The ant-tracking attribute effectively characterizes faults F2 and F3 (Fig. 13b), while fault F1 is difficult to observe. Anomalies in the ant-tracking attribute are found at the locations of Zones 3-1, 3-2, and 3-5, while no anomalies are observed in the other fracture zones.

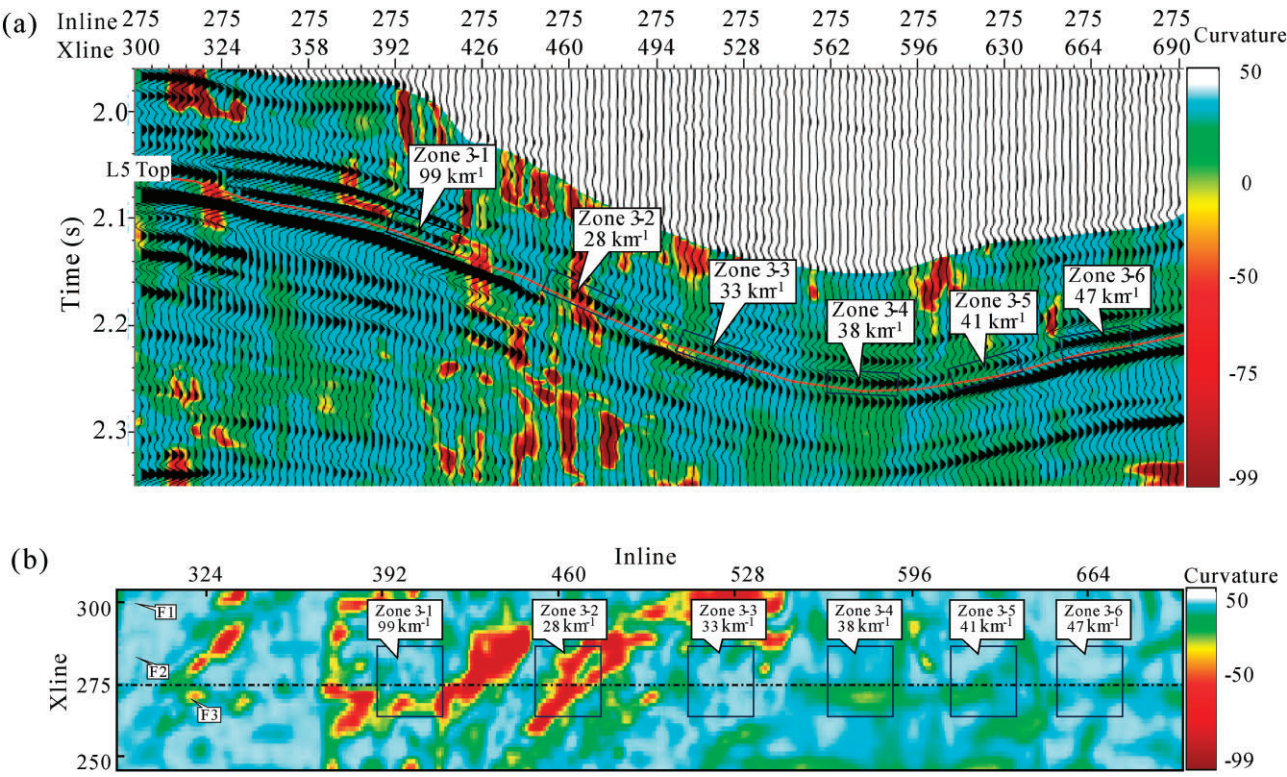


Figure 12. (a) Merged most negative curvature attribute and seismic profiles of Inline 275. (b) The most negative curvature slice along the top of layer L5. The black squares indicate fractured zones.

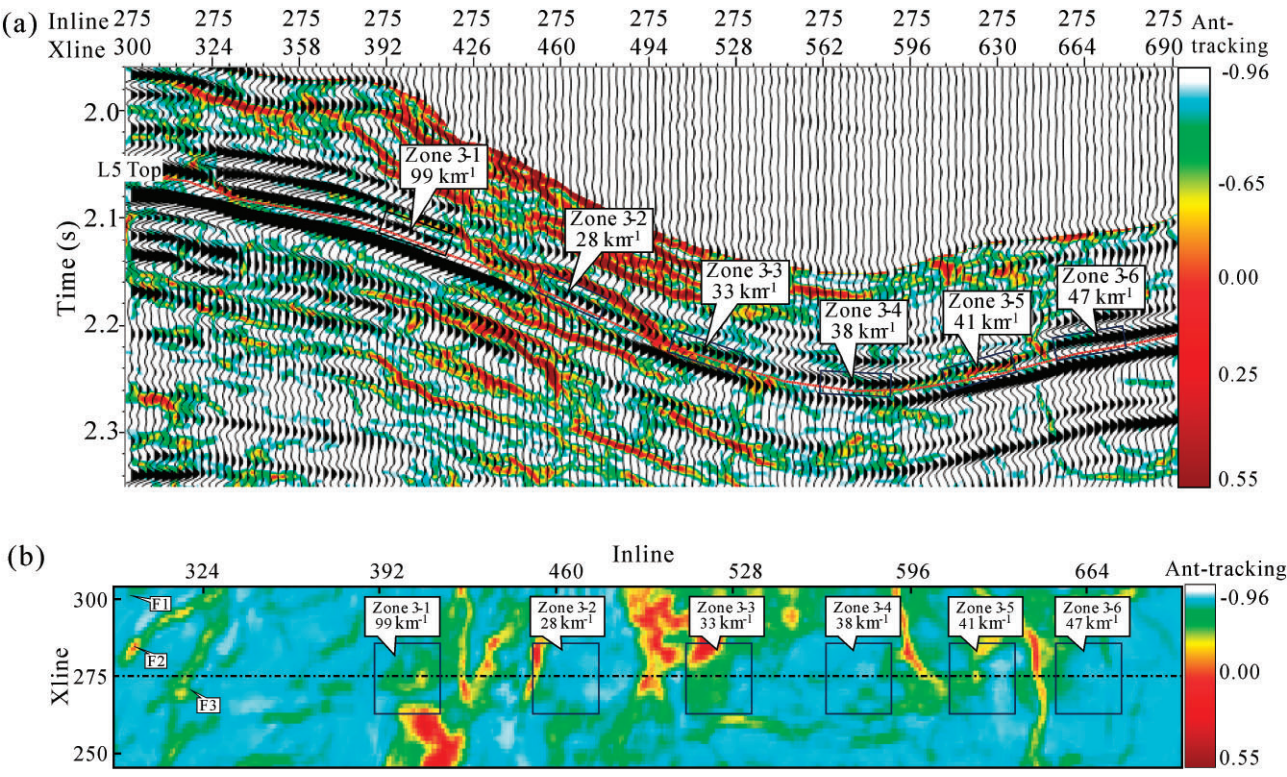


Figure 13. (a) Merged ant-tracking attribute and seismic profiles of Inline 275. (b) The ant-tracking slice along the top of layer L5. The black squares indicate fractured zones.

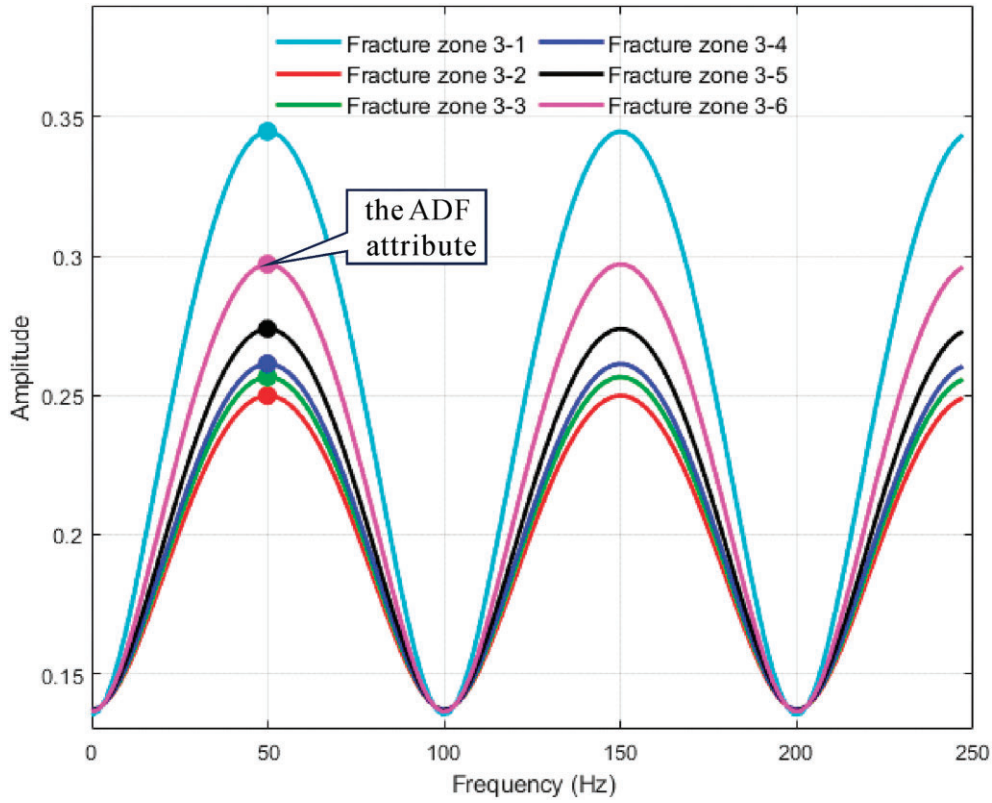


Figure 14. The short-window amplitude spectra of seismic reflection waveforms corresponding to six fractured zones. The ADF attribute of each fracture zone correlates with their fracture densities.

In summary, the seismic attributes of coherence, variance, most positive curvature, and ant-tracking demonstrate high sensitivity to faults, making them effective for characterizing macroscale fractures. However, these attributes exhibit limited sensitivity to mesoscale fractures, primarily due to the influence of small fracture scales, thin-bed tuning effects, and rapid variations in stratigraphic dip.

4. A frequency-dependent seismic attribute for mesoscale fracture density characterization

In this section, we introduce a frequency-dependent seismic attribute that is sensitive to the presence of mesoscale fractures in thin beds. This attribute, referred to as the ADF attribute within the first tuning period, can be obtained through spectral decomposition.

A time-domain seismic trace $d(t)$ can be transformed into the time-frequency domain using the short-time Fourier transform as follows (Sajid & Ghosh 2014):

$$D(\tau, f) = \int_{-\infty}^{\infty} d(t)w(t - \tau)e^{-i2\pi ft} dt, \quad (8)$$

where t is the time, f is the frequency, $d(t)$ is the seismic trace in the time domain, $w(t - \tau)$ is a convolutional window function, and τ is the translation along the time axis.

As to a thin-bed, the period of notches (P_f) of its seismic amplitude spectrum with respect to frequency (Hz) can be represented as follows (Partyka *et al.* 1999):

$$P_f = \frac{1}{\Delta T} = \frac{V_p}{2\Delta H}, \quad (9)$$

where ΔT is the two-way travel time interval (in seconds) of the thin bed, ΔH is the thickness (in meters) of the thin bed, and V_p represents the P-wave velocity (in m/s) within the thin bed. Thus, the ADF attribute can then be formulated as follows:

$$D\left(\tau, \frac{P_f}{2}\right) = \int_{-\infty}^{\infty} d(t)w(t - \tau)e^{-i2\pi \frac{P_f}{2}t} dt, \quad (10)$$

where $\frac{P_f}{2}$ refers to as the dominant frequency of the first notch period. The short-window amplitude spectra are calculated for the seismic reflection waveforms corresponding to the six fractured zones (Fig. 14). The P-wave velocity is 4000 m/s, and the equivalent thickness is 20 m. The notch period P_f is calculated using Equation (9), resulting in a value of 100 Hz, and the dominant frequency within the first notch period is 50 Hz. The ADF attributes (within the first notch period) are indicated in Fig. 14 and the ADF attribute increases with the fracture density.

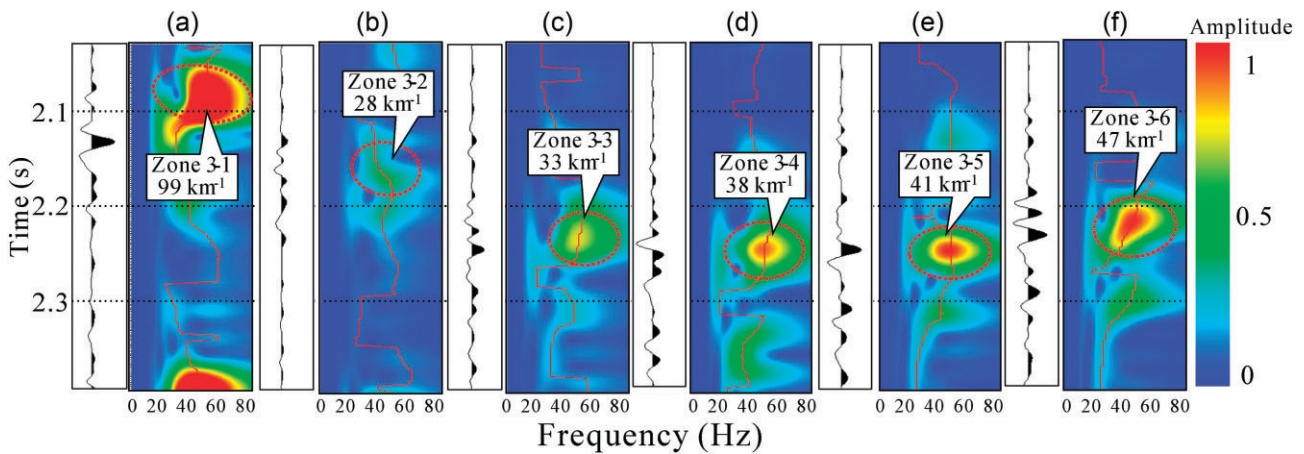


Figure 15. Time-frequency spectra of six seismic traces corresponding to six fracture zones: (a) Zone 3-1, (b) Zone 3-2, (c) Zone 3-3, (d) Zone 3-4, (e) Zone 3-5, and (f) Zone 3-6. Red ellipses indicate the locations of fractured zones.

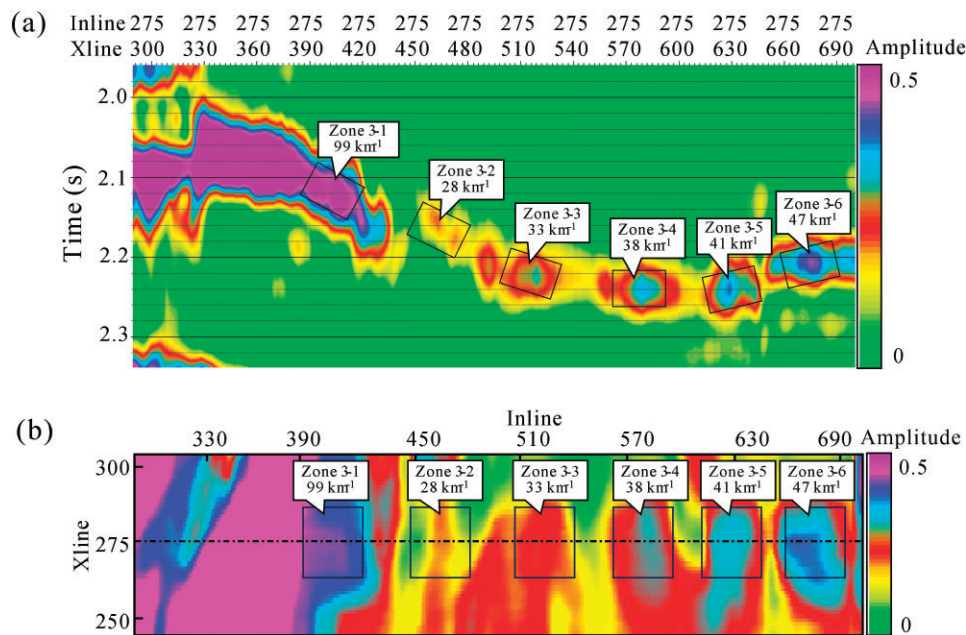


Figure 16. (a) The ADF attribute of Inline 275. (b) The ADF attribute horizon slice. Black squares indicate the locations of fracture zones.

Six seismic amplitude traces from the physical modeling seismic dataset, corresponding to six fracture zones, are decomposed into the time-frequency spectrum space, as shown in Fig. 15. Red ellipses indicate the locations of the fracture zones in the target layer L5. Figure 16 illustrates the dependence of fracture density on the ADF attribute within the target layer L5. Zone 3-1, characterized by a fracture density of 99 km^{-1} , represents the zone with the highest fracture density, with similar descriptions apply to Zones 3-2 through 3-6 for their respective fracture densities.

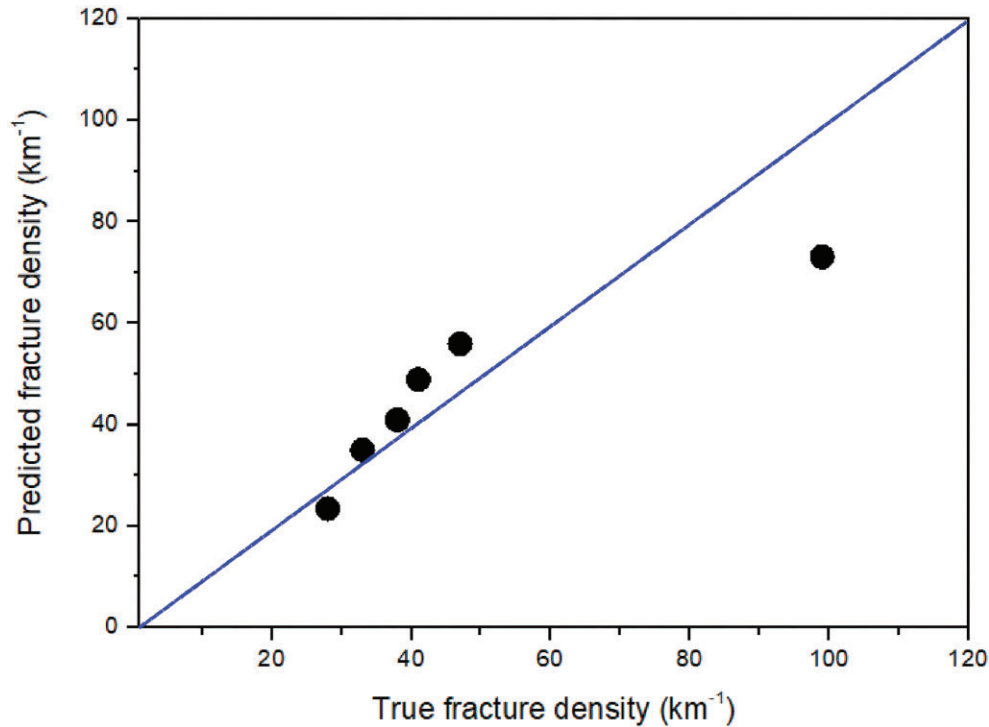
As observed in Figs 14–16, the dominant frequency within the first notch period is 50 Hz, and there is a quantitative relationship between the ADF attribute and fracture density. Zone 3-1, which exhibits the highest fracture den-

sity, corresponds to the most significant anomaly in the ADF attribute. From Zone 3-2 to 3-6, the fracture density values increase progressively, and the corresponding ADF attribute values also increase proportionally. Moreover, zones without fractures exhibit only smaller background values. Notably, the higher values in the area to the left of the Inline 275 profile (Xline range from 300 to 390) are attributed to the presence of faults in this area (Fig. 16).

To further validate the capability of the ADF attribute in quantitatively predicting fracture density, mean values corresponding to the fracture zones within the target layer L5 are extracted from the ADF attribute profile of the Inline 275. These values, presented in Table 3, are used to derive a relationship between fracture density and the ADF attribute.

Table 3. Average values derived from the ADF attribute profile of Inline 275 for the six fracture zones within the target layer L5, along with the fracture density prediction results.

Fracture zone	ADF attribute average value	True fracture density (km^{-1})	Predicted fracture density (km^{-1})
Zone 3-1	0.75	99.00	73.07
Zone 3-2	0.24	28.00	23.42
Zone 3-3	0.42	33.00	34.99
Zone 3-4	0.49	38.00	40.91
Zone 3-5	0.57	41.00	48.90
Zone 3-6	0.63	47.00	55.91

**Figure 17.** Crossplot between true fracture density and predicted fracture density for the six fracture zones.

The relationship is expressed as an exponential function, as follows:

$$\hat{y} = 13.71 \times e^{2.231x}, \quad (11)$$

where $\hat{y}$ is the predicted fracture density (km^{-1}) and x represents the average value of the ADF attribute.

The regression equation yields a correlation coefficient of 0.8067. Using Equation (11), fracture density predictions are made and are presented in Table 3. Figure 17 illustrates a crossplot comparing the true and predicted fracture densities of the six fracture zones. The prediction error increases with fracture density. This may be due to the increased scattering effect caused by fractures. The largest error occurs in Zone 3-1, primarily due to its proximity to a fault zone and its location within an interval with significant lateral variation. Therefore, geological complexities have a significant impact on fracture characterization using seismic data.

5. Conclusions

Using a physical modeling seismic dataset, we experimentally validated the effectiveness of several commonly used seismic attributes in characterizing multiscale fractures. Additionally, we introduced a frequency-dependent seismic attribute for characterizing mesoscale fractures in thin beds. Our findings can be summarized as follows:

- (i) Attributes such as coherence, variance, most positive curvature, and ant-tracking exhibit higher sensitivity toward faults and are effective for fault characterization. Furthermore, the amplitude attribute shows lower sensitivity to faults compared to other attributes.
- (ii) Most seismic attributes show limited sensitivity to mesoscale fractures, likely due to factors such as the small scale of fractures, thin-bed tuning effects, and rapid variation in stratigraphic dip.

- (iii) We introduce a frequency-dependent seismic attribute, the ADF within the first tuning period, which is effective for characterizing mesoscale fracture density in thin beds.

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Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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